Tobias Mwangi-Hartley

Data Scientist | Applied Machine Learning

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PROFILE

Data scientist with 6 years in analytics and data science, including 4+ years as a data scientist, building and applying machine learning models for forecasting, customer analytics and retention. Experienced in developing Python models and analytics workflows used by operations and growth teams, with production data workflows using BigQuery, dbt and Vertex AI Workbench. Strong statistical foundation and practical experience translating business questions into measurable analyses and decisions. Moving towards machine learning engineering.

EXPERIENCE

Data Scientist | Thistlebox (meal-kit subscriptions), Edinburgh *Feb 2022–Present*

- Built a weekly demand forecasting model for orders per recipe and region, forecasting up to six weeks ahead using LightGBM with feature engineering for calendar, price and menu-position features. WAPE decreased from 23% to 14%, with an estimated 9% reduction in food waste.
- Implemented a scheduled Google Cloud forecasting workflow in Vertex AI Workbench, writing model outputs to BigQuery for operational use in Looker.
- Developed an XGBoost churn propensity model, scored weekly in BigQuery, to target retention offers to top-decile customers. A 10% holdout validation found churn decreased by 2.1 percentage points in the treated group.
- Design and analyse approximately 40 A/B tests per year for growth and product teams; built a Python and Streamlit sequential testing calculator with sequential boundaries, used by around 25 colleagues.
- Wrote the calculator app and Dockerfile; the data platform team handles deployment, with the app hosted on Cloud Run.
- Write production SQL in BigQuery and maintain dbt models for the analytics layer.
- Present results to leadership monthly.

Junior Data Analyst | Firth Energy Retail, Glasgow *Sep 2020–Jan 2022*

- Conducted customer segmentation and tariff analysis using R and SQL.
- Automated monthly regulator reports using R Markdown.

SELECTED MACHINE LEARNING PROJECT

Image Classification Course Project | Personal *2025*

- Fine-tuned a small CNN in PyTorch on a public food-images dataset as part of an online deep-learning course; achieved approximately 80% accuracy. Learning project.
- Tools: PyTorch, torchvision, Google Colab.

EDUCATION

MSc Statistics | University of the Lothians *2020 | Distinction*

Dissertation: Bayesian hierarchical models for small-area energy demand

BSc Mathematics | University of the Lothians *2019*

TECHNICAL SKILLS

Programming and data: Python (pandas, scikit-learn, LightGBM, XGBoost, statsmodels), R (tidyverse, brms), SQL (BigQuery)

Machine learning and statistics: Feature engineering, time-series forecasting, churn modelling,

A/B testing, causal inference, Bayesian statistics

Tools and platforms: Google Cloud Platform (GCP): BigQuery, Vertex AI Workbench, Cloud Run; dbt, Looker, Streamlit, Docker, Git, Jupyter, PyTorch, torchvision

LANGUAGES

English (native) | Swahili (conversational)